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📅 Date: 10 March 2025

**Activity 3: What's on Your Plate? Unpacking Manpower, Budget, and Test Strategies**

**Objective:**
Students have identified several items on their plates based on the scenario, and now we will break them down in more detail. We will unpack these three foundational elements—**manpower**, **budget**, and **test strategies**—which will help students analyse and break them down by identifying their components and interdependencies.

**Note:** The timeline is crucial for this activity.

**Instructions:**

**Step 1: Individual Brainstorming (10 min)**
Each student will reflect on the three key aspects of a test project and **list the various elements that make up each one**.

- **Manpower:** What roles and skills are needed? How do team size and expertise impact the project?
- **Budget:** What are the cost factors? How does budget influence manpower and test strategies?
- **Test Strategies:** What different approaches can be used? How do they depend on manpower and budget?

Students should categorise their thoughts using the following structure:

- **Main Course (Core elements)** – The fundamental components of manpower, budget, and test strategies that directly impact the test project.
- **Side Dishes (Supporting factors)** – Additional considerations that influence decisions, such as constraints, stakeholder requirements, or risk factors.
- **Spices & Seasoning (Enhancements & refinements)** – Elements that improve efficiency, such as automation, resource optimisation, or contingency plans.
- **Leftovers (Out-of-scope items)** – Items that do not directly impact manpower, budget, or test strategies in this scenario.

**Step 2: Group Discussion – Unpacking the Three Elements (30 min)**
Students will form groups of 8 to 10 members and work together to **unpack** each of the three key elements, breaking them down into specific components.

**1. Manpower (Human Resources & Roles)**

- Roles & Responsibilities – for example, test managers, test analysts, automation engineers, developers, and end-users.

- Skills & Expertise - for example, manual testing, automation, performance testing, security testing.
- Availability & Allocation - for example, full-time vs. part-time testers, shared resources, outsourcing.
- Training & Competency Development - for example, onboarding, skill gaps, certifications.

**2. Budget (Cost & Resource Allocation)**

- Direct Costs - for example, salaries, tool licenses, testing infrastructure.
- Indirect Costs - for example, training, overhead costs, hardware/software upgrades.
- Budget Constraints & Trade-offs: Balancing cost vs. quality, prioritizing resources.
- Cost-Saving Strategies: Open-source tools, automation, outsourcing.

**3. Test Strategies (Approach & Execution Methods)**

- Types of Testing: for example, functional, non-functional, automation, exploratory.
- Tools & Frameworks: for example, selenium, JIRA, LoadRunner, Postman, CI/CD integration.
- Test Execution Planning: for example, manual vs. automated, risk-based, shift-left testing.

Each group will **identify the most critical elements for their project** and document their findings using a visual representation (whiteboard, flipchart, or digital tool).

**Step 3: Group Presentation (10 min per group)**

Each group will present their **breakdown of manpower, budget, and test strategies**, explaining:

- The key elements they identified for each category.
- Why they prioritised certain aspects over others.
- Any challenges or trade-offs they considered.

Groups should use a **diagram or structured list** to illustrate their findings.

**Step 4: Class Reflection & Lecturer Feedback (15 min)**

- The class will compare how different groups structured their manpower, budget, and test strategy breakdowns.
- The lecturer will provide insights into best practices for **managing manpower, budget constraints, and selecting effective test strategies** in real-world projects.

**Note:** Please refer to the provided scenario for this activity.